Zyphotek

Data Analytics Project Report

Title: South India College Market Research using Python and Power BI

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Date:30-6-2026

Objective:

The main objective of this project is to collect, organize, clean, analyze, and visualize data from colleges across South India. The project aims to build a comprehensive database containing details such as college information, courses offered, NAAC grades, placement statistics, and contact information of principals and placement officers. Using Python for data analysis and Power BI for visualization, the project generates meaningful insights that can support educational market research and business decision-making.

Methodology:

The project followed a systematic data analytics workflow consisting of the following phases:

- Data Collection
- Data Validation
- Data Cleaning
- Exploratory Data Analysis (EDA)
- Data Visualization
- Dashboard Development
- Business Insight Generation

Python libraries such as Pandas and Matplotlib were used for data processing and visualization, while Power BI was used to create an interactive dashboard.

Data Collection Process:

Data was collected from official college websites and other publicly available educational sources. The following information was gathered:

- College Name
- State and City
- Principal Details
- Placement Officer Details
- Contact Information
- Courses Offered
- Student Strength
- NAAC Grade
- Placement Statistics
- College Website

The collected information was stored in an Excel spreadsheet named Zyphotek.xlsx for further analysis.

1	S_No	Colllege_name	Location	Principal Name	P_C_Num	P_Email	Placement_officer_N	Placement_offic
2	1	Noorul Islam Centre for Higher Education	Kanyakumari	Dr.P.Thirumalcalavan	4651250566	registrar@	Dr. G. Aravind Swan	9442229791
3	2	Panimalar Engineering College	Chennai	Dr. K. Mani	4426490404	principal@	Mr. P. Balamurugan	044 - 26274665
4	3	St. Joseph's College of Engineering	Chennai	Dr. Vaddi Seshagiri F	44-24501060	jprstjosepl	Dr. N. Diwakar	99449 87560
5	4	Loyola College	Chennai	Rev. Dr. A. Louis Ar	44 28178301	principal@	Dr. M. Victor Louis /	9884925258
6	5	Madras Christian College	Chennai	Dr. P. Wilson	44-22390675	principal@	Dr. T. Robinson	44-22390675
7	6	Stella Maris College for Women	Chennai	Dr. Sr. Stella Mary	44-28110121	principal@	Ms. Sandra Mary	44-28113350
8	7	Ethiraj College for Women	Chennai	Dr. S. Uma Gowrie	44-28279189	principal@	Dr. J. Mangayarkaras	44-28279189
9	8	Presidency College	Chennai	Dr. R. Raman	44-28544894	presidency	Dr. S. Ramanathan	44-28544899
10	9	Women's Christian College	Chennai	Dr. Florence John	44-28275926	principal@	Ms. Annie Rachel	44-28275926
11	10	Queen Mary's College	Chennai	Dr. B. Uma Maheshw	44-28444995	qmcchenn	Dr. K. Sumathi	44-28444995
12	11	Justice Basheer Ahmed Sayeed College for	Chennai	Dr. Amthul Azeez	44-24349175	jbasprinci	Dr. Firdouse Jahan	44-24350395
13	12	PSG College of Technology	Coimbatore	Dr. K. Prakasan	422-2572265	principal@	Dr. R. Nadarajan	422-4344333
14	13	PSG College of Arts & Science	Coimbatore	Dr. D. Brindha	422-4303300	principal@	Dr. B. Ramesh	422-4303322
15	14	Sri Krishna Arts & Science College	Coimbatore	Dr. R. Jagajeevan	422-2678400	principal@	Mr. S. Suresh	9842851122
16	15	Karunya Institute of Technology & Scienc	Coimbatore	Dr. Elijah Blessing	422-2614300	ku@karun	Dr. D. Sujatha	422-2614536
17	16	SSN College of Engineering	Chennai	Dr. V. E. Annamalai	44 27469700	principal@	Dr. S. Radha	44 27469752
18	17	CIT (Coimbatore Inst. of Tech.)	Coimbatore	Dr. A. Rajeswari	422 2574071	principal@	Dr. K. Marimuthu	422 2573882
19	18	Thiagarajar College of Engg (TCE)	Madurai	Dr. M. Palaninatha R	452 2482240	principal@	Dr. G. Kothandaram	452 2482430

Additionaly data is been collected from students with the help of google forms to know their current status and the placement assistance they need. Which can be used to provide placement assistance based on their preferences.

Form_Responses

Timestamp	College Name	Student Name	Phone Number	Email ID	Academic Year	Department Name	Internship Status	Placement Attended Status	Pl
13/06/2026 16:55:28	Sri Sairam Engineering C	Balamurugan J	7418699463	lakshmibala2919@gmai	3rdYear	IOT	Not yet	No	
13/06/2026 16:58:43	PANIMALAR ENGINEERI	JEFRIN JA	8903733063	jefrinja05@gmail.com	3rdYear	CSE	Not yet	No	
13/06/2026 17:07:53	Sri Sairam Engineering C	YADAVA PRASATH C	7904124219	sec25cs14@sairamtap.e	3rdYear	CSE	Currently doing	No	
13/06/2026 18:29:53	Sri Sairam Engineering C	JANARTHANAN V	8778275267	janarthananv655@gmail	3rdYear	IOT	Not yet	No	
13/06/2026 18:40:59	Sri sairam engineering c	Aravinth I	6379835688	sec24ci011@sairamtap.	3rdYear	IOT	Currently doing	No	
15/06/2026 06:43:29	SRI SAI RAM ENGINEERI	S.DHANUSH	6381274591	dhanush.cseiot@gmail.c	3rdYear	IOT	Not yet	No	
20/06/2026 00:46:00	Sri sairam engineering c	H Pragadeshwaran	8870832870	pragadees.official01@gr	3rd Year	IOT	Not yet	No	
24/06/2026 16:38:58	MEPCO SCHLENK ENGC	K S SIVA KUMARAN	9578835127	sivakumarankannan111	3rd Year	MECH	Currently doing	No	
24/06/2026 16:41:29	Mepco schlenk engineer	Rohith	6369534920	roey2377@gmail.com	3rd Year	MECH	Completed	No	
24/06/2026 17:16:38	Mepco Schlenk Enginee	C.Rohith Kumar	8778957189	c rohithkumarsangeetha	3rd Year	MECH	Completed	No	
24/06/2026 18:58:39	SRI SAIRAM ENGINEERI	SRI DHATTATHREYA E	7305600212	sec24ci004@sairamtap.	3rd Year	IOT	Not yet	No	
24/06/2026 19:19:23	sri sairam engineering c	Vishwar S J	9342384995	sec24ci014@sairamtap.	3rd Year	IOT	Completed	No	
28/06/2026 16:46:48	University College of Enç	Ajay Neshanth T A	6382770544	ajayneshanth14@gmail.	3rd Year	ECE	Not yet	No	
28/06/2026 19:19:17	Sri Sai Ram Engineering	K SAI LINGESHWARAN	7338792956	sec24ci006@sairamtap.	3rd Year	IOT	Completed	No	
28/06/2026 19:52:17	Government College of T	Dhanuja A	7904529208	dhan.240371771062201	3rd Year	ECE	Completed	Yes	
28/06/2026 19:55:20	Government College of t	Kesavi M	9597350313	m.kesavi1308@gmail.cc	3rd Year	ECE	Completed	No	
28/06/2026 23:50:06	Government college of t	Malini R	7339686617	maliniravanan@gmail.cc	3rd Year	ECE	Completed	No	

Form responses 1

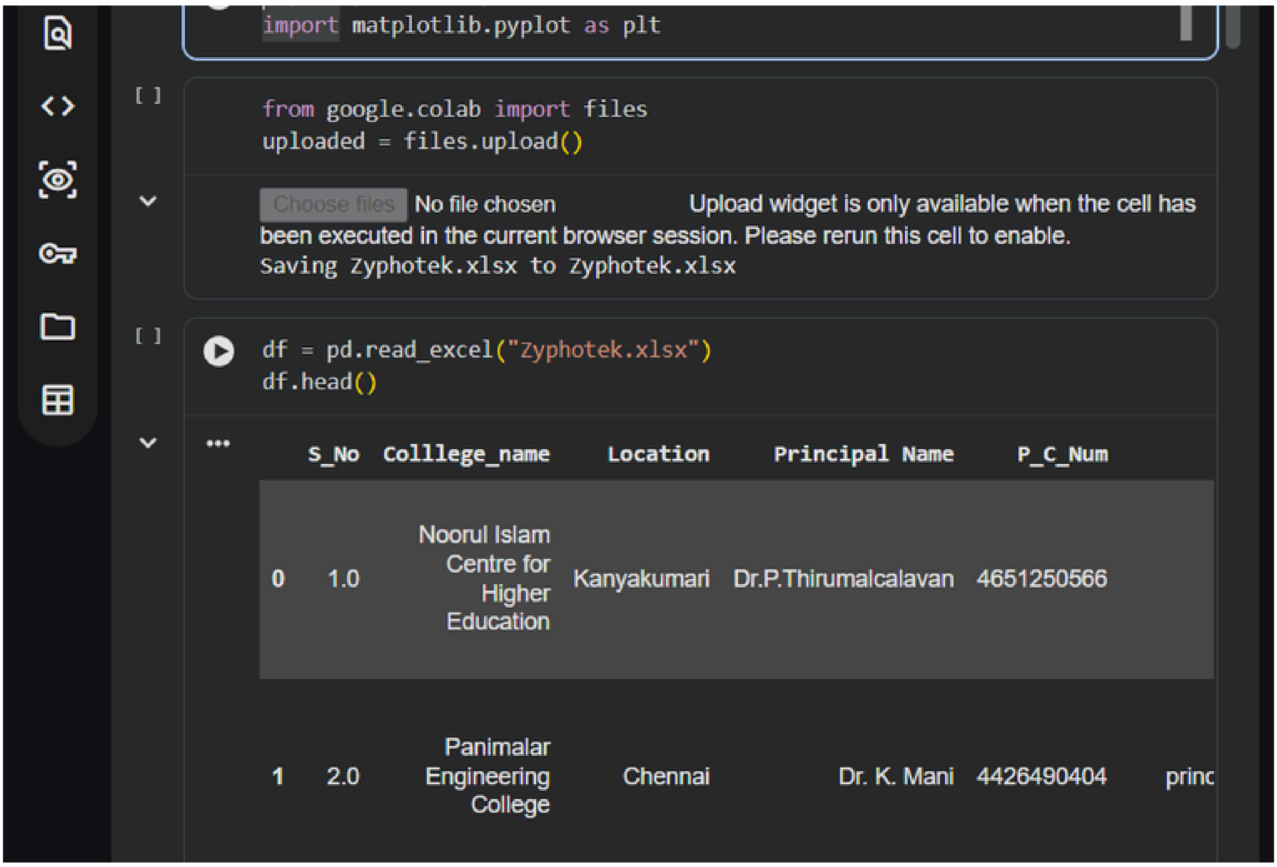
Data Cleaning Process:

Before analysis, the dataset was cleaned to improve data quality.

The following operations were performed:

- Removed duplicate records
- Handled missing values
- Corrected inconsistent text formatting
- Standardized state and college names
- Removed unnecessary spaces
- Verified important contact information

These preprocessing steps ensured that the dataset was accurate and suitable for analysis.

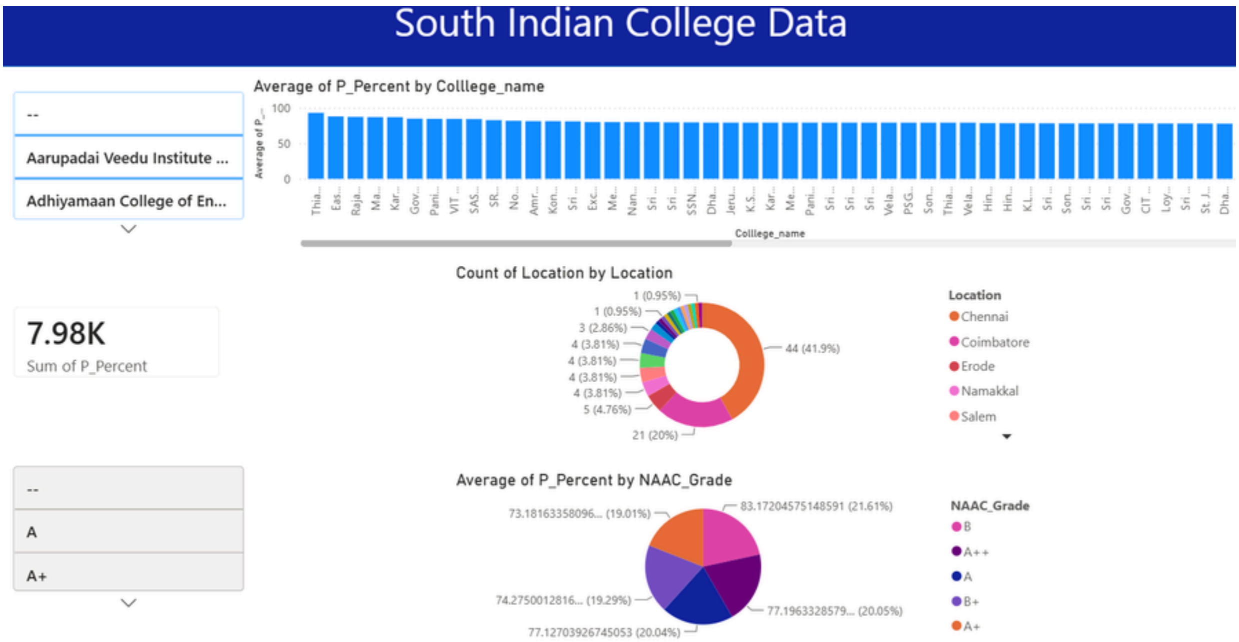


Key Insights:

After analyzing the dataset, several useful insights were obtained:

- Distribution of colleges across different states.
- Comparison between Government and Private colleges.
- Distribution of NAAC grades.
- Placement statistics across colleges.
- Identification of regions with a high concentration of educational institutions.

The Power BI dashboard presents these insights through interactive charts and graphs.



Recommendations:

Based on the analysis, the following recommendations are suggested:

- Update the database periodically to maintain accuracy.
- Verify contact information regularly.
- Include additional parameters such as NIRF ranking and accreditation status.
- Expand the database to cover colleges from other regions of South India the current dataset has more datas from chennai and coimbatore which is around 62% of data.

Future Scope:

The project can be enhanced by:

- Automating data collection using web scraping.
- Integrating SQL databases for large-scale storage.
- Building a web application using Flask for real-time dashboard access.
- Deploying the project on cloud platforms for wider accessibility.

Conclusion:

This project successfully demonstrates the complete Data Analytics lifecycle, from data collection and cleaning to analysis and visualization. Python enabled efficient data processing, while Power BI provided interactive dashboards for decision-making. The developed database serves as a valuable resource for educational market research and can be expanded further with automation.

THANK YOU!